

MARKET REGIME ANALYSIS ACROSS ASSET CLASSES: A UMAP CLUSTERING STUDY OF CRYPTOCURRENCY, FOREX, AND EQUITY MARKETS

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ABSTRACT. We present a comprehensive analysis of market regime structures across four major asset classes using UMAP [MHM18] dimensionality reduction and K-means clustering on intraday price patterns (2021-2024). Our analysis reveals fundamental structural differences between cryptocurrency markets (Bitcoin [Nak08]) and traditional markets (EUR/USD, AAPL, QQQ). Bitcoin exhibits distinct “trendy” regime-switching behavior with extreme asymmetry (75% bear days, 25% bull days) and excellent cluster separation (silhouette score 0.83), while all traditional markets show balanced, symmetric mean-reverting characteristics (50/50 split) with moderate cluster separation (0.55-0.59). Volatility follows a clear hierarchy: Bitcoin (1.05%) $\gg$ Equities (0.35-0.40%) $>$ Forex (0.18%).

1. INTRODUCTION

Understanding market regime dynamics is crucial for algorithmic trading strategies. While previous work has characterized individual markets, little research has systematically compared regime structures across asset classes using unsupervised learning techniques. This study addresses this gap by applying UMAP (Uniform Manifold Approximation and Projection) dimensionality reduction followed by K-means clustering to daily intraday patterns across cryptocurrency, forex, and equity markets.

Our key finding is that the “trendy” characterization often applied broadly to financial markets is, in fact, crypto-specific. Traditional markets (both stocks and forex) share remarkably similar structural properties despite their different underlying mechanisms, while cryptocurrency markets exhibit fundamentally different dynamics.

2. METHODOLOGY

2.1. Data Processing. We analyzed four major assets over the period 2021-01-01 to 2024-12-31:

- **Bitcoin (BTC):** X:BTCUSD, 1,460 complete days (24/7 trading)
- **Euro/US Dollar (EUR/USD):** C:EURUSD, 889 complete days (excludes weekends)
- **Apple Inc. (AAPL):** 998 complete days (NYSE hours 9:30-16:00 ET)
- **Invesco QQQ Trust (QQQ):** 998 complete days (NYSE hours 9:30-16:00 ET)

For each asset, we:

- (1) Obtained 15-minute bar data
- (2) Constructed daily vectors by normalizing each day to percentage changes from the opening price
- (3) For BTC (24/7): 96-dimensional vectors (96 15-minute intervals)
- (4) For EUR/USD (24/5): 96-dimensional vectors (filtered for 24-hour trading)
- (5) For stocks: 26-dimensional vectors (6.5 hours NYSE trading / 15 min = 26 intervals)

2.2. UMAP and Clustering. We applied UMAP dimensionality reduction with the following parameters:

- `n_neighbors` = 15
- `min_dist` = 0.1

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- `n_components = 2`
- `metric = 'euclidean'`
- `random_state = 42`

Following dimensionality reduction, we performed K-means clustering with $K=2$ through $K=5$ and evaluated cluster quality using silhouette scores. We selected $K=2$ for detailed analysis based on interpretability and silhouette score performance.

3. RESULTS

3.1. Bitcoin (BTC). Figure 1 shows the UMAP embedding and cluster assignments for Bitcoin.

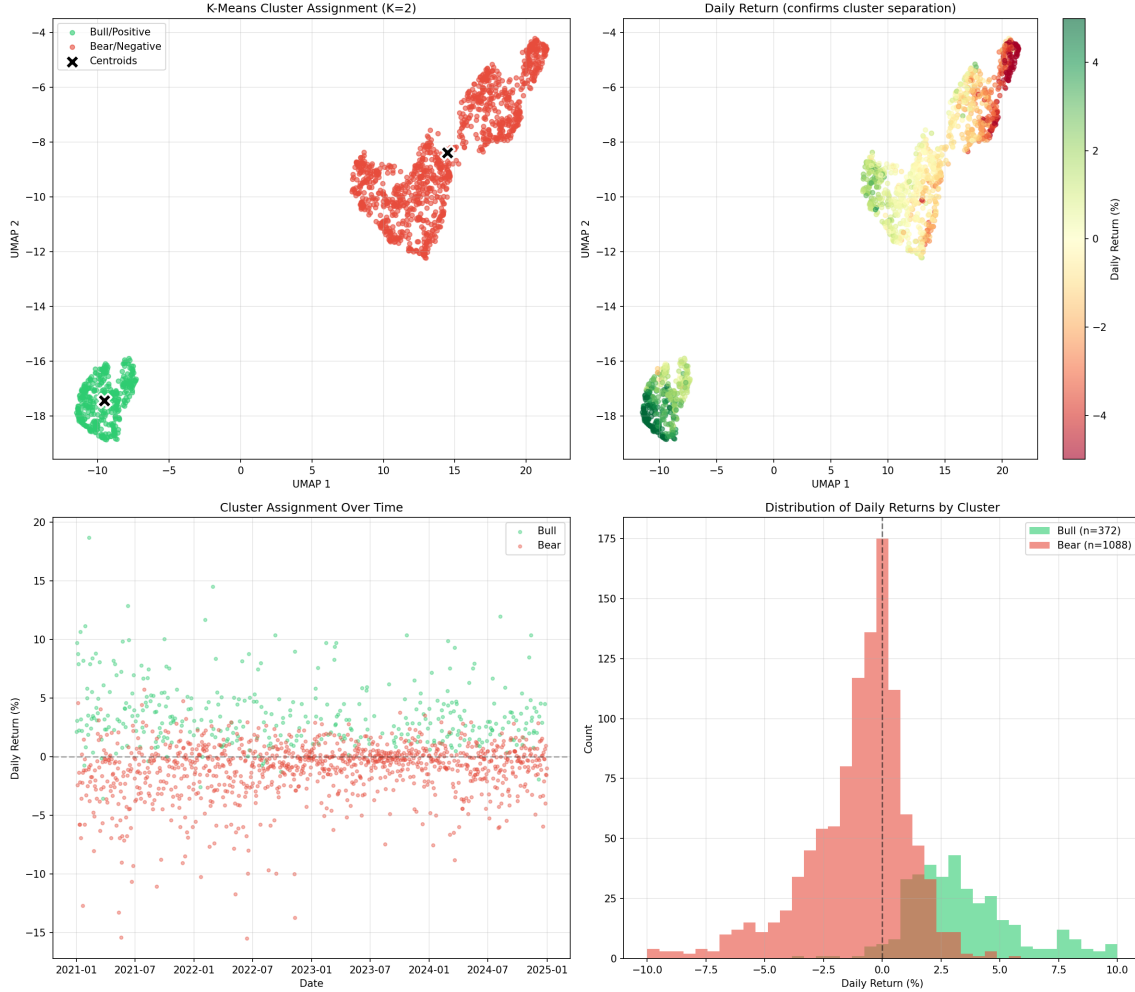


FIGURE 1. Bitcoin UMAP cluster analysis showing K-means assignments (top left), daily returns (top right), volatility patterns (bottom left), and temporal distribution (bottom right). Clear separation between bull and bear regimes is evident.

Cluster Quality: Silhouette score of 0.8274 indicates excellent separation between clusters.
Cluster Characteristics:

- **Bull/Positive Cluster (25.5% of days, 372 days):**
 - Average return: +3.62% (median +3.06%)
 - Volatility: 1.49%
 - Daily range: 5.59%
 - Return range: -3.55% to +18.69%
- **Bear/Negative Cluster (74.5% of days, 1,088 days):**

- Average return: -1.08% (median -0.53%)
- Volatility: 0.90%
- Daily range: 3.64%
- Return range: -15.51% to +5.75%

Temporal Dynamics:

- Total transitions: 560 over 4 years
- Average persistence: 2.6 days
- Median persistence: 1.0 day (very choppy)
- Longest regimes: Bear 25 days, Bull 4 days

Year-to-Year Variation: Bitcoin shows high variability in regime distribution across years, with 2023 being exceptionally bearish (81% bear days) and 2021 most bullish (33% bull days).

3.2. **EUR/USD (Forex).** Figure 2 presents the UMAP analysis for EUR/USD.

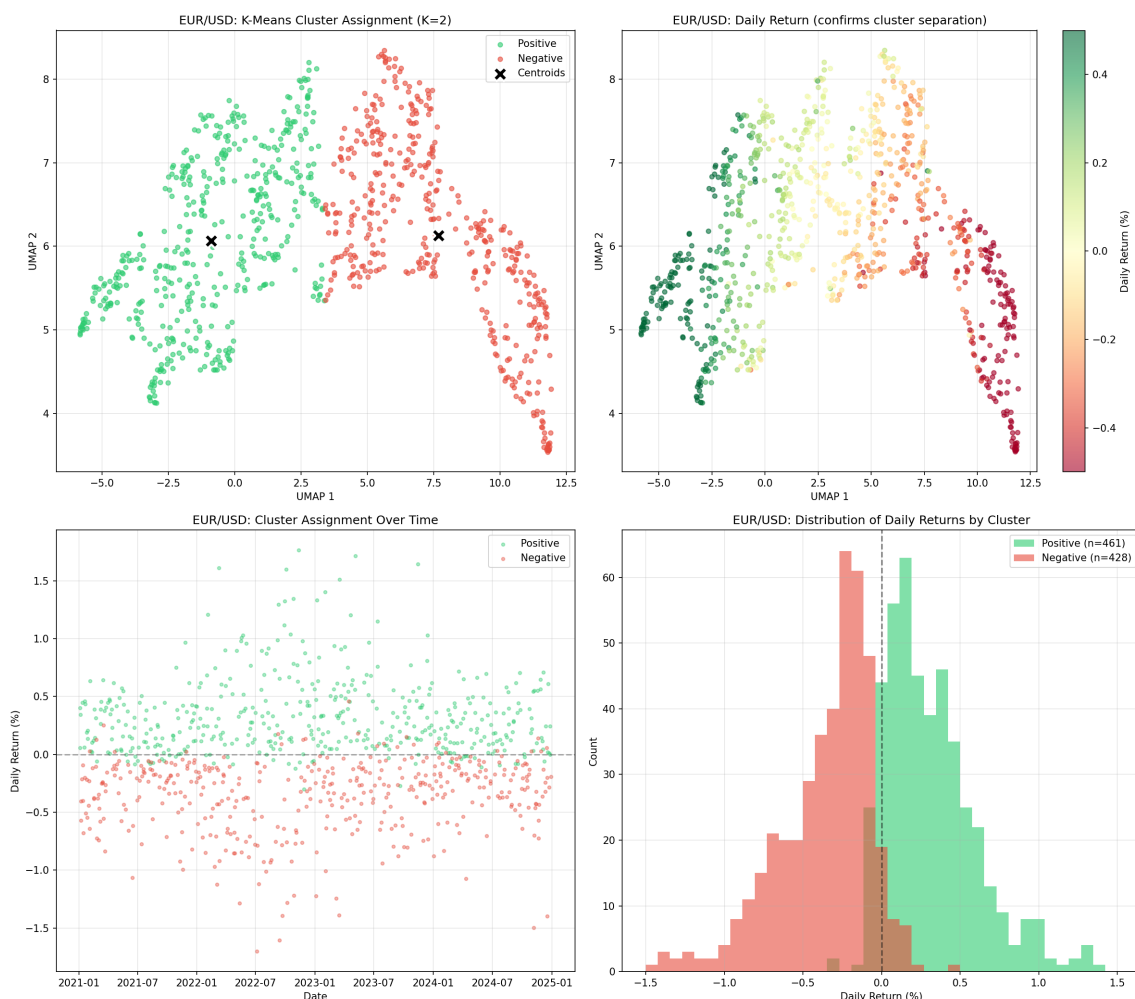


FIGURE 2. EUR/USD UMAP cluster analysis. Unlike Bitcoin, forex shows moderate cluster separation with substantial overlap and balanced positive/negative distribution.

Cluster Quality: Silhouette score of 0.5867 indicates moderate separation with significant overlap.

Cluster Characteristics:

- **Positive Cluster (51.9% of days, 461 days):**
 - Average return: +0.3289% (median +0.2566%)
 - Volatility: 0.1767%

- Daily range: 0.6258%
- **Negative Cluster (48.1% of days, 428 days):**
 - Average return: -0.3607% (median -0.2828%)
 - Volatility: 0.1873%
 - Daily range: 0.6495%

Temporal Dynamics:

- Total transitions: 452 over 4 years
- Average persistence: 2.0 days
- Median persistence: 2.0 days
- Longest regime: 9 days (both directions)

Year-to-Year Stability: Remarkably consistent 50-52% positive across all years (2021-2024).

3.3. Equity Markets (AAPL and QQQ). Figure 3 shows the UMAP analysis for both AAPL and QQQ.

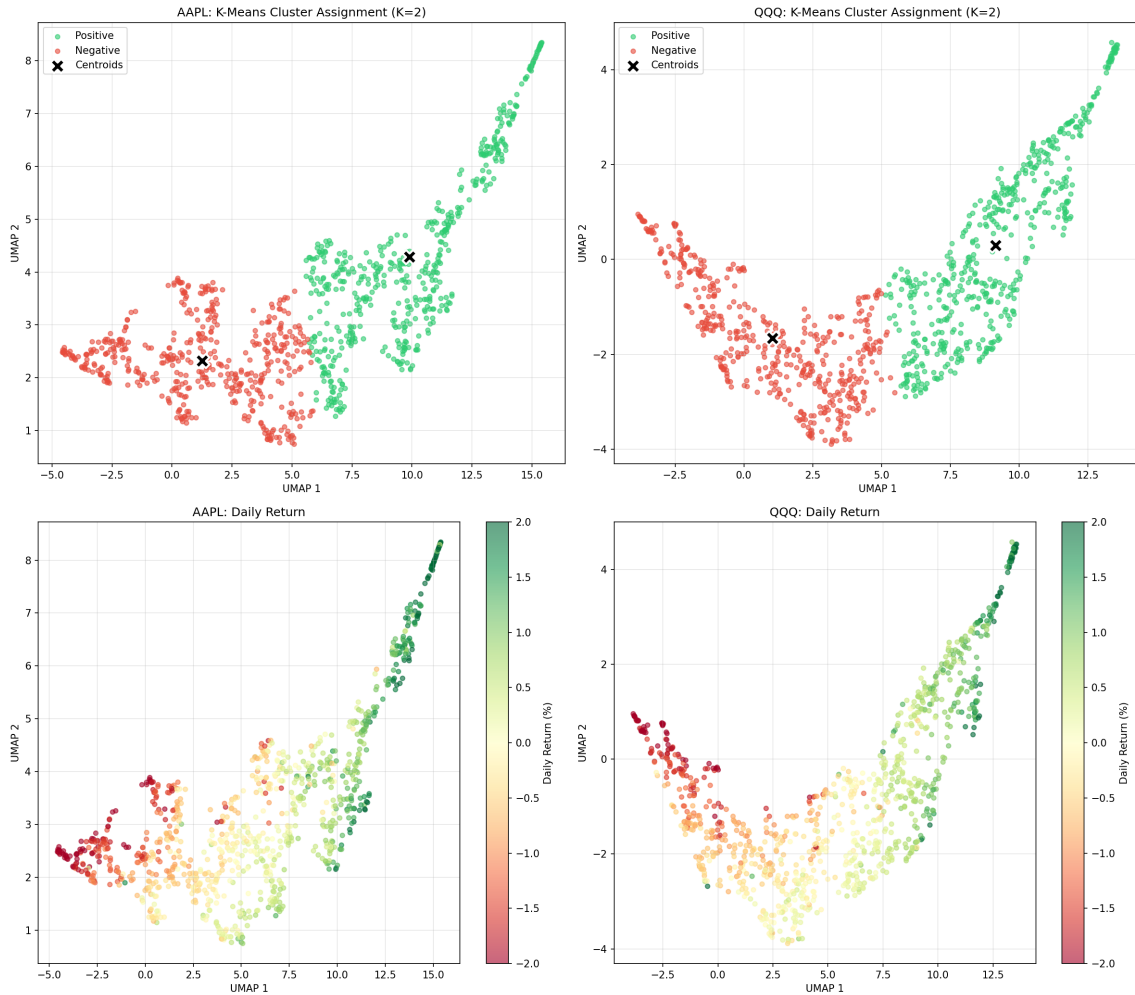


FIGURE 3. UMAP cluster analysis for AAPL (left) and QQQ (right). Both equity markets exhibit structure similar to EUR/USD: moderate separation, balanced distribution, and intermediate volatility between crypto and forex.

3.3.1. Apple Inc. (AAPL). Cluster Quality: Silhouette score of 0.5483.

Cluster Characteristics:

- Positive Cluster: 51.5% (514 days), +0.8679% avg return

- Negative Cluster: 48.5% (484 days), -0.7907% avg return
- Average volatility: 0.40%
- Median persistence: 2.0 days, longest regime 10 days

3.3.2. *Invesco QQQ Trust (QQQ)*. **Cluster Quality:** Silhouette score of 0.5758 (highest among traditional markets).

Cluster Characteristics:

- Positive Cluster: 51.6% (515 days), +0.7791% avg return
- Negative Cluster: 48.4% (483 days), -0.7706% avg return
- Average volatility: 0.35%
- Median persistence: 2.0 days, longest regime 11 days

4. COMPARATIVE ANALYSIS

4.1. **Cluster Separation.** Bitcoin demonstrates significantly better cluster separation (silhouette 0.8274) compared to all traditional markets (0.55-0.59), indicating that BTC has clear regime boundaries while traditional markets have fuzzy regimes with substantial overlap.

4.2. **Asymmetry. Bitcoin** exhibits highly asymmetric structure:

- 75% bear days (small losses) vs 25% bull days (large gains)
- Bull days return 3.4 $\times$ more than bear days lose
- Extreme tail events (± 15 -19%)

Traditional Markets are nearly symmetric:

- EUR/USD: 52% / 48% (+0.33% / -0.36%)
- AAPL: 51.5% / 48.5% (+0.87% / -0.79%)
- QQQ: 51.6% / 48.4% (+0.78% / -0.77%)
- Limited tail events (± 2 %)

This suggests Bitcoin has “lottery ticket” dynamics while traditional markets are mean-reverting.

4.3. **Volatility Hierarchy.** Table 1 summarizes key metrics across all assets.

TABLE 1. Cross-Asset Comparison of Market Regime Characteristics

Metric	BTC	QQQ	EUR/USD	AAPL	BTC vs Trad.
Silhouette Score	0.8274	0.5758	0.5867	0.5483	1.4-1.5 $\times$
Avg Daily Vol. (%)	1.05	0.35	0.18	0.40	3-6 $\times$
Cluster Balance	25/75	52/48	52/48	51/49	Extreme asym.
Avg Return (+) (%)	+3.62	+0.78	+0.33	+0.87	4-11 $\times$
Avg Return (-) (%)	-1.08	-0.77	-0.36	-0.79	1.4-3 $\times$
Med. Persistence (d)	1.0	2.0	2.0	2.0	0.5 $\times$ (choppy)
Longest Regime (d)	25	11	9	10	2.3-2.8 $\times$
Total Transitions	560	485	452	475	1.2 $\times$
Year-Year Variance	High	Low	Low	Low	Variable vs Stable

Volatility follows a clear hierarchy: BTC (1.05%) $\gg$ Stocks (0.35-0.40%) > Forex (0.18%). Bitcoin is approximately 3 $\times$ more volatile than equities and 6 $\times$ more volatile than forex.

4.4. **Regime Persistence. Bitcoin** exhibits unstable switching:

- Median 1 day (flips daily)
- Can have extended bear runs (25 days)
- Bull runs are short-lived (max 4 days)

Traditional Markets show stable persistence:

- All have median 2-day persistence

- Maximum regimes 9-11 days
- No extreme asymmetric persistence

This indicates BTC regimes are harder to predict (rapid daily switching) but recognizing them is more valuable (larger return differences). Traditional markets show stable, predictable 2-day persistence.

4.5. Year-to-Year Stability. Bitcoin shows high variability (18.9% to 33.2% bull days across years), while all traditional markets are remarkably stable (50-52% positive every year). This suggests BTC market character changes significantly year-to-year, while traditional markets are structurally consistent.

5. TRADING STRATEGY IMPLICATIONS

5.1. Why 8-Month Lookback Benefits BTC. Our backtest results show that an 8-month rolling window (240-day lookback) achieves +5.29% cumulative return vs -7.65% for a 4-month window, an advantage of +12.94 percentage points. This validates our UMAP findings:

- (1) **Regime Recognition:** Longer history helps identify which cluster (bull/bear) is currently active
- (2) **Asymmetry Learning:** Model needs extensive data to learn the 75/25 split and asymmetric returns
- (3) **Regime Transitions:** 8 months captures 90 regime changes, teaching transition patterns
- (4) **Year Variability:** Longer window captures different market characters (2021 vs 2023 mix)

5.2. Traditional Markets Require Different Approaches. For EUR/USD, AAPL, and QQQ:

- (1) **Stable Structure:** 50/50 split is constant across years
- (2) **Symmetric Returns:** No complex asymmetry to learn
- (3) **Shorter Regimes:** 2-day median persistence fits well in shorter windows
- (4) **Lower Volatility:** Smaller moves require less historical context

Similar strategies could work across these assets given their structural similarities. The emphasis should be on technical mean-reversion rather than regime detection.

6. CONCLUSIONS

This study reveals fundamental structural differences between cryptocurrency and traditional markets:

- (1) **Crypto markets (BTC) are fundamentally different from ALL traditional markets:** Higher volatility (3-6 $\times$), asymmetric regime structure (75% bear, 25% bull vs 50/50), extreme returns in bull regime (+3.62% avg vs +0.33-0.87%), variable market character year-to-year vs stable, and much higher cluster separation (0.83 vs 0.55-0.59).
- (2) **The “trendy” characterization is CRYPTO-SPECIFIC:** BTC shows clear regime-switching with extended bear runs (25 days max). ALL traditional markets (EUR/USD, AAPL, QQQ) are balanced, symmetric, and mean-reverting. Stocks show intermediate volatility between BTC and forex, closer to forex, however.
- (3) **Traditional markets likely require different modeling approaches:** Shorter lookback may suffice given stable 50/50 structure. Focus should be on technical mean-reversion rather than regime detection. Similar strategies could work across EUR/USD, AAPL, QQQ given their structural similarities.

These findings have important implications for algorithmic trading strategy design: cryptocurrency markets require sophisticated regime detection with extended lookback periods, while traditional markets benefit from simpler mean-reversion approaches with shorter windows.

6.1. Future Research Directions.

- (1) Test optimal lookback windows for other cryptocurrencies (ETH, etc.) and forex pairs
- (2) Develop explicit regime detection beyond implicit model learning
- (3) Investigate regime transition prediction signals
- (4) Explore multi-asset portfolios combining trendy crypto with mean-reverting traditional markets
- (5) Apply alternative clustering approaches (HDBSCAN, temporal clustering)

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